Lesson-23

ACOUSTIC PHONETICS-I

At the end of this section, the students will be able to:

- EXPLAIN and DEFINE acoustic phonetics, source/filter theory and tube models.

Topic-114: Acoustic Phonetics

As an important part of phonetics, acoustics is the study of the physics of the speech signal quiz (i.e., when speech sound travels through the air in a wave form) from the speaker's mouth to the hearer's ear through vibrations. In acoustics, it is possible to measure and analyze these vibrations (physical properties) by mathematical techniques usually by using specially-developed computer software to produce spectrograms of speech. Acoustic phonetics also studies the relationship between activity in the speaker's vocal tract and the resulting sounds by involving physics, computer, statistics and a number of experiments in lab. Thus, the analysis of speech using the expertise available in acoustic phonetics is claimed to be more objective and scientific than the traditional auditory method which mostly depends on the reliability of the trained human ear. It also involves advanced level speech software for analyzing

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Topic-115: Explaining Acoustic Phonetics

Acoustics is a branch of phonetics which studies the physical properties of speech sound as transmitted between the mouth of a speaker and the ear of a listener. It is mainly devoted to the principles of physics involving the latest technology such as computer and software (e.g., Praat). It is the scientific study of speech sounds wholly dependent on the use of instrumental (lab based) techniques of investigation. It particularly involves electronics and some grounding in physics; however, mathematics is a prerequisite for the advanced study of the subject. In this course, we are going to learn mainly about the basic features of speech sounds (both consonants and vowels). Starting from the very beginning, we will particularly see the fundamental experiments of acoustics (such as recording of speech and annotating and ultimately distinguishing among various forms and features of speech sounds. This course will also introduce you to the mostly used software in acoustic phonetics (i.e., Praat) and its application in the phonetic analysis of human speech. Based on these introductory level experiments during this course, it is expected that the learners would be able to plan advanced level applications and experiments of acoustics.

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Topic-116: Acoustic Analysis

quiz

According to the experts of speech sounds (phoneticians), acoustic analysis can provide a clear, objective datum for the investigation of speech – the physical 'facts' of utterance. In other words, acoustic evidence is often referred to when one wants to support an analysis being made in articulatory or auditory phonetics. However, it is important to note down that one should not be too reliant on acoustic analyses

which are subject to many mechanical limitations (e.g. the need to calibrate measuring devices

accurately), and which are often themselves open to multiple interpretations. Acoustic analysis not only gives us the features of a sound but also tells us about the duration or length of a speech sound. For such an analysis, we need to carefully know about material and procedure of recording. Thus, acoustic analysis describes the durational characteristics, articulatory properties and phonetic differences through physiological measurement.

Topic-117: Acoustic Analysis of Vowels

The experts of phonetics are particularly interested in analyzing vowels acoustically. They describe vowels in terms of numbers (in a language that how many vowels are possible). It is also possible to analyze vowel sounds so that the measurement of the actual frequencies of the formants (the formant structure of the vowels of a language) is taken. Having taken these formants of vowel sounds, they can be represented graphically (plotting them on chart) as given for English vowels in your textbook. The figure (given in your textbook) gives the average of a number of authorities' values of the frequencies of the first three formants in eight American English vowels. The first three formants of the eight vowels of the American English words *heed, hid, head, had, hod, hawed, hood, who'd* have been taken. Explore the acoustic features of these vowels as recorded by the authors (Ladefoged and Johnson).

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Topic-118: Source Filter Theory of Speech Production important quiz and s.q.

Source-filter theory is an important concept in acoustic phonetics. It is a model of speech (e.g., vowel) production. According to this theory, source refers to the waveform of the vibrating larynx. Its spectrum is rich in harmonics, which gradually decrease in amplitude as their frequency increases. The various resonance chambers of the vocal tract, especially the movements of the tongue and lips, act on the laryngeal source in the manner of a filter (see filtered speech), reinforcing certain harmonics relative to others. Thus the combination of these two elements (larynx as source and cavity as filter) is known as the source-filter model of speech (e.g., vowel) production. We have already discussed that speech sounds can

Topic-118: Source Filter Theory of Speech Production important quiz and s.q.

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Topic-119: Explaining Source – Filter Mechanism

In this theory, the tract is represented using a source-filter model and several devices have been devised to synthesize speech in this way. The idea is that the air in the vocal tract acts like the air in an organ pipe, or in a bottle. Sound travels from a noise-making source (i.e., the vocal fold vibration) to the

some of the sound energy reflects back into the vocal tract. The addition of the reflected sound energy with the source energy tends to amplify energy at some frequencies and damp energy at others, depending on the length and shape of the vocal tract. The vocal folds (at larynx) are then a source of sound energy, and the cavity (vocal tract - due to the interaction of the reflected sound waves in it) is a frequency filter altering the timbre of the vocal fold sound. This idea can make it very easy for us to understand the formants of a vowel sound. Thus this same source-filter mechanism is at work in many musical instruments. In the brass instruments, for example, the noise source is the vibrating lips in the mouthpiece of the instrument, and the filter is provided by the long brass tube.

Further readings on this section:

- Chapter 8 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)
- Online sources: http://www.phon.ox.ac.uk/jcoleman/tubes_practical.html

Lesson-24

ACOUSTIC PHONETICS-II

At the end of this section, the students will be able to:

- EXPLAIN and DEFINE perturbation theory and the acoustics of consonants.

Topic-120: Tube Models

Now in order to fully understand the concept of vowels and their formants, we introduce another idea which is the tube model of vocal tract. Actually, the formants that characterize different vowels are the result of the different shapes of the vocal tract. Any particle of air, such as that in the vocal tract or that in a bottle, will vibrate in a way that depends on its size and shape. Remember that the air in the vocal tract is set in vibration by the action of the vocal folds (in larynx). Every time the vocal folds open and close, there is a pulse of acoustic energy (activation). Irrespective of the rate of vibration at source (of the vocal folds), the air in the vocal tract will resonate at these frequencies as long as the position of the vocal organs remains the same. Because of the complex shape of the filter (tract), the air will vibrate in more than one way at once. So, the relationship between resonant frequencies and vocal tract shape is actually much more complicated than the air in the back part of the vocal tract vibrating in one way and the air in other parts vibrating in another. Here we will just remember the fact that in most voiced sounds, three formants are produced every time the vocal folds (source) vibrate. Note an interesting point here that the resonance in the vocal tract (filter) is independent of the rate of vibration of the vocal folds (source). In other words, the vocal folds may vibrate faster or slower, giving the sound a higher or lower pitch, but the formants will be the same as long as the position of the tube (vocal tract) is the same. This point is further explained in the next topic.

Topic-121: Explaining the Tube Models

quiz + s.q

Actually this is an old idea (initiated by the great German scientist Hermann Helmholtz) that a vowel is merely the rapid repetition of its peculiar note. Now it is said that a vowel is the rapid repetition (corresponding to the vibrations of the vocal folds) of its peculiar two or three notes (first two or three formants respectively). Actually, all voiced sounds are distinguishable from one another by their formant structure (frequencies). This idea could be understood by considering the vocal tract as a tube and thus the concept is when the vocal fold pulses have been produced at a steady rate, the "utterance" is on a monotone. In other words, what you hear as the changes in pitch are actually the changes in the overtones of this monotone "voice." These overtone pitch variations convey a great deal of the quality of the voiced sounds. The rhythm of the sentence is apparent because the overtone pitches occur only when the vocal folds would have been vibrating. This tube model allows us to get an idea towards understanding of resonance, resonant frequencies (formants), the source-filter model and some other aspects of the relationship between articulatory configurations (of vocal tract) and their acoustic consequences.

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Topic-123: Explaining Perturbation Theory

According to perturbation theory, resonance occurs in a uniform tube where one end is closed and the other end is open (just remember the source filter idea). This theory tells us whether each resonance frequency increases or decreases when a small modification occurs in the diameter at a local region of the tube (tract). As a result:

-the resonance frequency of the particular resonance mode decreases when a constriction is located at an anti-node of that resonance mode; and
-the resonance frequency of the particular resonance mode increases when a constriction is located at a node of that resonance mode.

Keeping in mind this idea of perturbation theory, we can derive that the resonance frequencies will change (decrease or increase) as per the position (modification in the size and nature) of the vocal tract (tube).

Topic-124: Explaining Acoustic Analysis (Vowels)

Using computer programs, we can analyze vowel sounds by showing their components through the display (spectrogram). In spectrograms, time runs from left to right, the frequency of the components is shown on the vertical scale, and the intensity of each component is shown by the degree of darkness. It is thus a display that shows, roughly speaking, dark bands for concentrations of energy at particular frequencies—showing the source and filter characteristics of speech. Remember that the traditional

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Topic-125: Acoustics of Consonants

The acoustic properties (structure) of consonantal sounds are usually more complicated than that of vowels. Usually, a consonant can be said to be a particular way of beginning or ending a vowel sound because during the production of a consonant there is no distinguishing feature prominently visible. There is virtually no difference in the sounds during the actual closures of voiced stops [b, d, g], and absolutely none during the closures of voiceless stops [p, t, k], because there is only silence at these points. Each of the stop sounds conveys its quality by its effect on the adjacent vowel. We have seen that during a vowel such as [u], there will be formants corresponding to the particular shape of the vocal tract. In the case of consonants, these changes are not really distinguishable (particularly for obstruents). Although there are some consonantal sounds which have vowel like structure; therefore, their acoustic features are somehow similar to vowels (in the case of nasal consonants, approximants and glides) but most of the consonants have totally different acoustic features. These differences in the nature of consonant sounds give us a pretty easy opportunity to distinguish consonants from vowels (e.g., in the case of stops, the complete closure is easily visible as there is just a simple thin line on the spectrogram (explore some spectrogram of consonant sounds and confirm this feature).

In order to analyze the acoustic features of consonants, use the instructions given in Chapter 8 of your course book as rough guides. Also remember that particular combinations (of various sounds) are important as neighboring vowels help a lot in guessing the exact consonant sounds. There is a list of hints and ideas given in the next topic to predict consonantal sounds on the basis of their acoustic features.

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Further readings on this section:

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- Online sources: http://www.splab.net/APD/G400/index-e.html
  http://www.phon.ox.ac.uk/jcoleman/tubes_practical.html

ACOUSTIC PHONETICS-III

At the end of this section, the students will be able to:

- EXPLAIN and INTERPRET spectrograms of speech sounds and think about some individual differences.

Topic-126: Explaining the Acoustics of Consonants

In addition to the above given features of consonant sounds (in Topic-125), following hints and ideas are important to be remembered for interpreting the spectrograms of consonants:

Very important for exams

Voiced - vertical striations corresponding to the vibrations of the vocal folds
Bilabial - locus of both second and third formants comparatively low
Alveolar - locus of second formant about 1700–1800 Hz.
Velar - usually high locus of the second formant
Retroflex - general lowering of the third and fourth formants
Stop - gap in pattern (with burst for voiceless and sharp formant beginning for voiced stops)
Fricative - random noise pattern in higher frequency regions
Nasal - formant structure similar to that of vowels (with formants at 250, 2500, and 3250)
Lateral - formant structure similar to that of vowels (with formants at 250, 1200, and 2400)
Approximant - formant structure similar to that in vowels, usually changing

Topic-127: Interpreting Spectrograms

Words recorded in isolation are easy to be interpreted through spectrogram but interpreting the spectrogram of connected speech is certainly a more difficult challenge. But the question is: Why should the students of phonetics and phonology learn spectrogram? So, before learning how to interpret spectrograms, the importance of learning and using spectrogram is given here:

1. Using Praat (or any other software) and spectrogram is particularly useful when a researcher is working on a problem related to the nature (physical properties) of a sound (e.g., is it a phoneme or allophone?).
2. It increases our understanding of the speech sounds and their behavior in different forms (in isolation or as the part of connected speech).
3. Practice on spectrogram gives us the opportunity to learn about the characteristics of speech sounds.
4. It is also important for experts who are working on phonetic aspects of speech as signal processing.
5. These are also used as the part of techniques in speech recognition.

6. Spectrograms enable us to explore the complex nature of speech structure (as the part of spoken language).
7. Spectrograms are the part of techniques used in machine translation.

In the light of the above points (and many more indeed), it is advisable to learn these techniques related to spectrograms and increase your expertise as there are various resources available online for the purpose.

Topic-128: Useful Techniques for Interpreting Spectrograms

There are certain useful points to be remembered for interpreting segments in the connected speech. We can start exploring them by looking at the segments one by one. In addition to the points mentioned above (in Topic-126), following are the helpful points to use in interpreting spectrograms:

1. Start analyzing sounds one by one by keeping in mind the individual characteristics (as given in Topic-126) of sounds as a class.
2. Carefully see the overall structure, especially the frequency scale.
3. While interpreting consonants, also analyze the behavior of the adjacent vowels.
4. Pay more attention to the first two formants (especially for vowels).
5. Watch for a burst and aspiration in stop sounds.
6. Remember that in vowels the first formant is inversely related to the height of a vowel (the lower

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it from a typical speech community). We also need to discount, at times, the purely individual features of linguistically significant speech. Moreover, for other particular reasons (such as used in forensic investigations), we need to discuss the validity of the identification of a particular speaker. Now the question is how to study individual variation. Following points are important in this regard:

1. Spectrograms show relative quality (e.g., a particular speaker may have a higher level of vowels).
2. The formant plots (of an average speaker) may be compared with the formant plots of a particular speaker.
3. Similarly, when two different speakers record their sets of vowels with the same phonetic quality, their relative positions on a formant chart will be similar, but the absolute values of the formant frequencies will differ.

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4. In such case, the absolute values of the vowels will be important. But remember that it is a

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5. One very important strategy may be to use the values of F4 (which may be the indicator of individual head size). Thus F4 may be studied in connection with first three formants for further evaluation. Remember that F4 for other sounds will also be required for such a comparison.

Topic-130: Analyzing Individual Differences

Further on individual differences, as an alternate method, a complete range of a speaker's vowel qualities may be considered as representative of the speaker's personal features which, in turn, may be compared with the formant frequency of each vowel (with the total range of that formant in that speaker's voice). But this is true that the phoneticians are still working with comparing the acoustic data of one individual with the other and improve further the system of speech recognition. Experts of applied phonetics and computer speech technology are trying to understand the complexity of speech – synthesis systems and improve it. Areas such as intonation and rhythm are being further explored for the purpose. For correct estimates, they are considering working with syntax, pragmatics, rhythmic structure and individual segmental influences together to develop computer programs capable to be used for speech recognition. But it is also true that such topics are the areas of computer engineers but, yes, phoneticians

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individual segmental influences together to develop computer programs capable to be used for speech recognition. But it is also true that such topics are the areas of computer engineers but, yes, phoneticians (as well as speech therapist and the experts of hearing sciences) can also play an important role in it. Secondly, speech identification is also directly needed by the legal experts for their court proceedings and 'voice-prints' are now largely used by lawyers. Spectrograms, although with a limited application at this stage, are potential area for future research studies.

Further readings on this section:

- Chapter 8 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)

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Lesson-26

VOWELS AND VOWEL-LIKE ARTICULATIONS-I

At the end of this section, the students will be able to:

- DEFINE and EXPLAIN vowels and vowel-like articulations with the help of primary and secondary cardinal vowel systems.

Topic-131: Vowels and Vowel-like Articulation

The fundamental distinction between consonant and vowel sounds is that vowels make the least obstruction to the flow of air. In addition to this, vowels are almost always found at the center of a syllable, and it is very rare to find any sound, other than a vowel which can stand alone as a whole syllable. Phonetically, each vowel has a number of features (properties) that distinguish it from other vowels. These include: firstly, the shape of the lips (lip-rounding), rounded (for sounds like /u/ vowel)

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various languages of the world (e.g., nasality – whether a vowel is nasal or not).

Topic-132: Cardinal Vowels

In order to classify vowels (independent of the vowel system of a particular language), the English phonetician Daniel Jones introduced a system in early 20th century and worked out on a set of vowels called the "cardinal vowels" comprising of eight vowels to be used as reference points (so that other vowels could be related to them like the corners and sides of a map). Jones' idea of cardinal vowels became a success and it is still used by experts and students for vowel description. He was strongly influenced by the French phonetician Paul Passy, and it has been claimed that the set of 'cardinal vowels' is quite similar to the vowels of educated Parisian French of his time. Cardinal vowel system is a chart or four-sided figure (the exact shape of which has been changed from time to time), with eight corners as can be seen on the IPA chart from IPA website. It is a diagram to be used both for rounded and unrounded vowels, and Jones proposed that there should be a primary and a secondary set of cardinal vowels. The primary set includes eight vowels in total (from 1 to 8); the front unrounded vowels [i, e, ɛ, a], the back unrounded vowel [ɑ] and the rounded back vowels [ɔ, o, u].

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Topic-133: Secondary Cardinal Vowels

Further on explaining vowels (particularly, the vowels of Non-European languages), a set of secondary cardinal vowels (as a precise set of references) was introduced by the same British phonetician Danial Jones (1881-1967). Secondary cardinal vowels are easy to understand in connection with the primary cardinal vowel system. The main difference between primary and secondary cardinal vowels is related to lip-rounding as in some languages the feature of lip-rounding is possible for front vowels. By reversing the lip position (in comparison with primary cardinal vowels), the secondary series of vowel types is produced (e.g., rounding the lips for the front vowels). In addition to the primary cardinal vowels (from vowel 1 to 8), following are the secondary cardinal vowels (their numerical codes and features) as pointed out Danial Jones:

9. Close (high) front rounded vowel	[y]
10. Close-mid front rounded vowel	[ø]
11. Open-mid front rounded vowel	[œ]
12. Open (low) front rounded vowel	[œ]
13. Open (low) back rounded vowel	[ɒ]
14. Open-mid back unrounded vowel	[ʌ]
15. Close-mid back unrounded vowel	[ɤ]
16. Close (high) back unrounded vowel	[ɯ]

There are two more vowels representing the highest possible position at the center of the tongue

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Lesson-27

VOWELS AND VOWEL-LIKE ARTICULATIONS-II

At the end of this section, the students will be able to:

- DEFINE and EXPLAIN various types of vowels in different languages (including British English and other accents of English).

Topic-135: Vowels in Other Accents of English

Vowel sounds in various accents of English are interesting for a number of reasons. They provide solid basis for comparisons and contrasts. Researchers have explored this aspect in various Englishes (the term has been frequently used in recent times) and have published descriptions of the auditory quality of the vowels in a number of accents of English language. Spectral structure of these vowels (by referring to the average formant frequencies of vowel systems in these accents) provides interesting comparisons. For example, the accent of American (newscasters) English has represented fairly conservative difference (in the first two formants) with Californian English. The Californian English does not maintain a contrast between the vowels in cot and caught (they are both spoken as the same [ɑ]). Moreover, the Californians have a higher vowel (lower first formant) in [ɛɪ] than in [ɪ]. Their high back vowels seem more towards front as they have a higher second formant. Among other differences, vowel /ʊ/ is often pronounced with spread lips in this variety of English. Similarly, in a number of northern cities in the United States (e.g., Pittsburgh and Detroit), [æ] is spoken very close to [ɛ] (as raised with decreased F1). These are some of

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While discussing vowels in English, we can explore vowels in BBC accent for a number of reasons. The first and the foremost comparison may be with the American English (or varieties of American English). Both of the varieties have been discussed by linguists and politicians alike (from both sides). The vowels of BBC accent are different in both number (20 in total – both pure vowels and diphthongs) and quality. British English speakers distinguish the vowel [ʌ] in cut from the vowel [ɜ] in curt. It does not have any r-coloring (rhotacization) mainly by the frequency of the first formant. Moreover, as the main feature of BBC English (to be noted here) is the distinction between the three back vowels [a] as in father and cart, [ɑ] as in bother and cot, and [ɔ] as in author and caught. BBC vowels are really interesting in making it a standard variety of English.

Topic-137: Vowels in Other Languages

Vowels in languages other than English are important to be explored here in connection with the on-going discussion. Variations among human languages are not limited only to the number of vowels but also to the quality and features. The vowel systems of some of the major languages are briefly mentioned here. Spanish has a very simple system contrasting only five vowel sounds [i, e, a, o, u]. It is important to

remember that these symbols do not have the same values as English or their descriptions of cardinal vowels. Japanese also has a set of five vowels [i, e, a, o, u] but very different in a narrower transcription than that of Spanish. Similarly, Danish has different vowels and their qualities as it contrasts three front rounded vowels which may occur both in long or short forms. Focusing on Pakistani regional languages, Urdu and Punjabi have very different nasal vowels (the nasality is phonemic here). Urdu has some seven nasal vowels such as /he/ vs. /hẽ/ or /hi:/ vs. /hĩ:/ as in word /nahĩ:/ meaning 'no'. Nasalization in vowels is a common feature of Indo Aryan languages.

Topic-138: Advanced Tongue Root (ATR)

Normally, while discussing the degree of variation in vowel sounds, three types of features are given (i.e., height of tongue, backness of tongue and lip rounding) which cover the major variation in world's languages. But this description does not cover all types of variation in vowel quality. One out of such variations is advanced tongue root (ATR) which is found in Akan language spoken in Ghana. Actually, vowels produced with ATR involve the furthest-back part of the tongue, opposite to the pharyngeal wall, which is not normally involved in the production of speech sounds - also called the radix (articulations of this type may, therefore, be described as radical). ATR (a kind of articulation in which the movement of the root of tongue expands the front–back diameter of the pharynx) is used phonologically in Akan (and some other African languages) as a factor in contrast of vowel harmony. The opposite direction of movement is called retracted tongue root (RTR). ATR is thus related to the size of pharynx – making the pharyngeal cavity different: creating comparatively large (+ATR: root forward and larynx lowered) and small pharyngeal cavity (-ATR: no advanced tongue root). Akan contrasts between two sets of vowels +ATR and –ATR.

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Topic-139: Rhotacized Vowels

In the description of vowel quality, rhotacization (or rhotacized vowel) is a term which is used in English phonology referring to dialects or accents where /r/ is pronounced following a vowel, as in words 'car' and 'cart'. Thus varieties of English are divided on the basis of this feature - varieties having this feature are rhotic (in which /r/ is found in all phonological contexts) while others (not having this feature) are non-rhotic (such as Received Pronunciation where /r/ is only found before vowels as in 'red' and 'around'). Similarly, vowels which occur after retroflex consonants are sometimes called rhotacized vowels (they display rhotacization). It is important to mention that while BBC pronunciation is non-rhotic, many accents of the British Isles are rhotic, including most of the south and west of England, much of Wales, and all of Scotland and Ireland. Most American English speakers speak with a rhotic accent, but there are non-rhotic areas (e.g., the Boston area, lower-class of New York and the Deep South).

Lesson-28

VOWELS AND VOWEL-LIKE ARTICULATIONS-III

At the end of this section, the students will be able to:

- DEFINE and EXPLAIN nasalization, semivowels and secondary articulatory gestures involved with vowels.

Topic-140: Nasalization in Vowels

In the description of vowels, nasalization is another feature which is not covered by the traditional method of description as it is not found in all languages of the world. The speakers of Urdu, Punjabi and many other Pakistani regional languages learn to produce a variety of nasal vowels as the part of their mother tongue and face no issue in learning nasalization in vowels. However, for the speakers of other languages (such as English which does not have nasal vowels) have to learn this feature of vowels by starting saying the low vowel /æ/ as in man and by keeping the soft palate lowered. Many languages have contrasts between nasal and oral vowels including French and Urdu. Urdu and Punjabi have many nasal vowels. Urdu has seven nasal vowels such as /he/ (meaning 'is') vs. /hẽ/ (meaning 'are'). Nasalization in vowels is a common feature of Indo Aryan languages. In IPA chart, the diacritic used for nasalization is

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Topic-141: Summary of Vowel Quality

In this course, we have discussed many features related to the description of vowel sounds. In this topic, we provide a summary of all these features related to vowels. There are two features of vowel quality (i.e., height and backness of the tongue) that are used to contrast one vowel with another in nearly all languages of the world. But there are four other features that are used less frequently and not all languages exhibit them. They include 'lip-rounding', rhotacization, nasalization and advanced tongue root (ATR). The following are the acoustic correlates of all these six features:

- Height - frequency of formant one (inversely related to F1)
- Backness - difference between frequencies of F2 and F1
- Rhotacization - frequency of formant three
- Rounding - lip position (rounded, half rounded or neutral)
- ATR - width of the pharynx (ATR or RTR)
- Nasalization - position of the soft palate

Topic-142: Semivowels

Most of the world languages contain a class of sounds that functions in a way similar to consonants but is phonetically similar to vowels (e.g., in English, /w/ and /j/ as in 'wet' and 'yet'). When they are used in the first part of syllables (at onset), they function as consonants. But if they are

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lists a semivowel corresponding to the back close unrounded vowel /ɯ/. Like the others, this is classed as an approximant.

Topic-143: Secondary Articulatory Gestures (SAG)

While discussing vowels, we need to highlight some of the secondary articulation related to vowels. Remember that 'secondary' articulation is an articulatory gesture with a lesser degree of closure occurring at approximately the same time as another (primary) gesture. It is different than co-articulation which is at the same time and of the same value (taking place as an equal level gesture). Thus it is appropriate to consider four types of secondary articulations in conjunction with vowels because they can usually be described as added vowel-like articulations including; 'palatalization' (adding a high front tongue gesture as in sound /i/), velarization (raising of the back of the tongue), pharyngealization (it is the superimposition of the narrowing of the larynx) and labialization (the addition of lip-rounding).

Topic-144: SAGs Discussion

The four types of possible secondary articulatory gestures related to vowel quality are described here for understanding and further detailed exploration. Palatalization is the addition of a high front tongue gesture (as in sound like [i]) to another main (primary) gesture. The diacritic used for palatalization is the small [ʲ] superimposed above another symbol (for primary gesture). The terms palatalization (a process whereby the place of an articulation is shifted nearer to the center of the hard palate) and palatalized (when the front of the tongue is raised close to the palate while an articulatory closure is made at another point in the vocal tract) are sometimes used in a slightly different way. A

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Velarization involves raising the back of the tongue (adding the /u/ vowel like quality). It can be considered as the addition of an [u]-like tongue position (but remember that it is without the addition of the lip rounding). A typical English example of velarization is the /l/ sound at the end of a syllable (as in words like kill, pill, sell and will) called velarized or dark /l/ and may be written as [ɫ]. The diacritics for velarization are both [ˠ] and [̴].

Another secondary gesture of vowel quality is pharyngealization which is the superimposition of narrowing of the pharynx. The IPA diacritics for symbolizing pharyngealization are [ ] (as for velarization) and [ˤ] (the superimposition of the symbol for pharyngeal sound).

The last secondary articulatory gesture for vowel quality to be noted here is labialization which is

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Topic-145: Summary of the SAGs

The secondary gestures for vowel quality are hereby summarized for further clarification. A sound may or may not have any of the four secondary articulations such palatalization, velarization, pharyngealization and labialization. Labialization may also have (at the same time) any of the three secondary articulatory gestures (and even if the sound is itself labial such as [mʷ]). The main features of these secondary articulatory gestures for vowel sounds are briefly mentioned here for your understanding:

1. Palatalization [ʲ] is the raising of the front of the tongue such as for /i/ vowel).
2. Velarization (written as [ˠ] and [˞]) is the raising of the back of the tongue (such as [u]-like sound.
3. Pharyngealization [ˤ] is the retracting of the root of the tongue.
4. Labialization [ʷ] is the rounding of the lips such as Arabic [sʷ] and [tʷ].

Further readings on this section:

- Chapter 9 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)

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Lesson-29

SUPRASEGMENTAL FEATURES-I

At the end of this section, the students will be able to:

- DEFINE, EXAMINE and EVALUATE suprasegmental features such as syllables and stress.

Topic-146: Introduction to Suprasegmental (SS) Features

Supra means 'above' or 'beyond' and segments are sounds (phonemes). Suprasegmental is a term used in phonetics and phonology to refer to a vocal effect (such as tone, intonation, stress, etc.) which extends over more than one sound (segment) in an utterance. Major suprasegmental features include pitch, stress, tone, intonation or juncture. Remember that these features are meaningful when they are applied above segmental level (on more than one segment). Phonological studies can be divided into two fields: segmental phonology and suprasegmental phonology. Suprasegmental features have been extensively explored in the recent decades and many theories have been constituted related to the application and description of these features.

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seg-men-tal. Phonetically, we can observe that the flow of speech typically consists of an alternation between vowel-like states (where the vocal tract is comparatively open and unobstructed) and consonant-like states where some obstruction to the airflow is made (thus altering speech between the two natural kinds of sounds). So, from the speech production point of view, a syllable consists of a movement from a constricted or silent state to a vowel-like state and then back to constricted or silent state. From the acoustic point of view, this means that the speech signal shows a series of peaks of energy corresponding to vowel-like states separated by troughs of lower energy (sonority).

Topic-148: Explaining Syllable

Phonologists are interested in the structure of a syllable. It can be divided into three possible parts as phonemes may occur at the beginning (onset), in the middle (nucleus or peak) and at the end (coda) of syllables - the combination of nucleus (peak) and coda is called the rhyme. The beginning (onset) and ending (coda) are optional while a syllable must have a nucleus (at least one phoneme). Thus, the study of the sequences of phonemes is called phonotactics, and it seems that the phonotactic possibilities of a language are determined by its syllabic structure (sequences of sounds that a native speaker produces can be broken down into syllables). Syllables are claimed to be the most basic units in speech (in a language – a sound system) meaning that an utterance must contain at least one syllable. Similarly, every language

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Topic-147: Syllable

The 'syllable' is a fundamentally important unit both in the fields of phonetics and phonology. Experts normally keep the phonetic notions of the syllable separate from the phonological ones. It is easy to understand but very difficult to define the syllable. In a simple way of defining the term, syllables are the parts of word (in which a word is further divided into parts), for example, *mi-ni-mi-za-tion* or *sup-re-seg-men-tal*. Phonetically, we can observe that the flow of speech typically consists of an alternation between vowel-like states (where the vocal tract is comparatively open and unobstructed) and consonant-like states where some obstruction to the airflow is made (thus altering speech between the two natural kinds of sounds). So, from the speech production point of view, a syllable consists of a movement from a constricted or silent state to a vowel-like state and then back to constricted or silent state. From the acoustic point of view, this means that the speech signal shows a series of peaks of energy corresponding

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has syllables, and children learn to produce syllables before they can manage to say a word of their native language. It is important to note that syllable structure could be of three types: 'simple' (CV), 'moderate'

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has syllables, and children learn to produce syllables before they can manage to say a word of their native language. It is important to note that syllable structure could be of three types: 'simple' (CV), 'moderate' (CVC) and 'complex' (with consonant clusters at edges) such as CCVCC and CCCVCC (where V means vowel and C stands for consonant). Moreover, words can have one syllable (monosyllabic), two syllables (bisyllabic or disyllabic), three syllables (trisyllabic) or many syllables (polysyllabic).

Topic-149: Stress as a Suprasegmental Feature

Stress is basically a suprasegmental feature applied to a whole syllable when it is made prominent by adding factors such as loudness, rise in pitch, length of duration and vowel quality (in contrast with other syllables). For example, in mi-ni-mi-**ZA**-tion, the second last (penultimate) syllable is prominent (as greater amount of energy is applied to it) as it is louder and longer than the rest of the syllables in the word. Similarly, its quality and pitch are different than that of others. All these factors make it stressed (compare: **IN**-sult vs. in**SULT** and be-**LOW** vs. **BILL**-ow). Stress is an important feature both in phonetics and phonology. Despite the fact that it has been extensively studied, there remain many areas of disagreement among the experts. To begin with a basic point, it is almost certainly true that in all languages some syllables are in some sense stronger than other syllables; these are syllables that have the potential to be described as stressed. Stress plays an important role in conveying (and changing) meaning.

Topic-150: Types and Categories of Stress

One of the areas in which a little agreement is found is related to the levels of stress. Some descriptions of languages manage with just two levels (stressed and unstressed), while others use more than two. In English, one can argue that if one takes the word 'in-di-ca-tor' as an example, the first

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Topic-151: Lexical and Emphatic Stress

In terms of its linguistic function, stress is often treated under two different headings: word (lexical) stress and sentence (emphatic) stress. Lexical stress is basically related to the primary stress applied at syllable level (when only one syllable is stressed) that has the ability to change the meaning and

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role in intonation patterns and rhythmic features of the language showing specific emphasis on the stressed word (which may be highlighting some information in the typical context). In order to perceive the nature of sentence level stress, read the following sentences with shifting the stress accordingly and judge the shift in emphasis (and its role in the context):

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Lesson-30

SUPRASEGMENTAL FEATURES-II

At the end of this section, the students will be able to:

- Define syllable-timed and stress-timed languages.

Topic-152: Stress Timed Languages

It is sometimes claimed by the experts that different languages (and dialects) have different types of rhythmic patterns. Languages of the world are; therefore, divided into two broad categories: stress timed language and syllable timed languages. But this is basically the division of languages on the basis of their modes of timing (i.e., stress vs. syllable timed languages). Stress timed languages have stress as their dominating rhythmic feature meaning that these languages seem to be timed according to the stressed patterns (the division among the syllables is made on the basis of stressed and unstressed patterns e.g., English and German languages). In other words, in stress timed languages, stressed syllables occur with regular intervals and their units of timing are perceived accordingly. Stress-timed rhythm is one of these rhythmical types, and is said to be characterized by a tendency for stressed syllables to occur at equal intervals of time. This idea is further clarified in the next topic.

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Topic-153: Explaining Stress Timed Languages

'Stress timed languages' is a very general phrase used in phonetics to characterize the pronunciation of languages displaying a particular type of rhythmic pattern that is opposed to that of syllable-timed languages. In stress-timed languages, it is claimed that the stressed syllables recur at regular intervals of time (stress-timing) regardless of the number of intervening unstressed syllables as in English. This characteristic is sometimes also referred to as 'isochronism', or isochrony. However, it is clear that this regularity is the case only under certain conditions, and the extent to which the tendency towards regularity in English is similar to that in, say, other Germanic languages remains unclear. In short, the division among the syllables is made on the basis of stress and unstressed patterns. In such languages, stress is realized both at word and sentence levels approximately changing the rhythmic patterns (particularly at sentence level).

Topic-154: Syllable Timed Languages

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syllables tend to occur at regular intervals of time with fixed word stress. A classic example is Japanese in which all morae have approximately the same duration. This tendency is contrasted with stress-timing where the time between stressed syllables is said to tend to be equal irrespective of the number of unstressed syllables in between. Czech, Polish, Swahili and Romance languages (e.g., Spanish and

French) are often claimed to be syllable-timed. Many phoneticians; however, doubt whether any language is truly syllable-timed.

Topic-155: Explaining Syllable Timed Languages

Syllable timed is a very general term used in phonetics to characterize the pronunciation of languages displaying a particular type of rhythmic pattern. It is opposed to stress-timed languages. In syllable-timed languages, the syllables are said to occur at regular intervals of time, (e.g., in French) this

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Topic-155: Explaining Syllable Timed Languages

Syllable timed is a very general term used in phonetics to characterize the pronunciation of languages displaying a particular type of rhythmic pattern. It is opposed to stress-timed languages. In syllable-timed languages, the syllables are said to occur at regular intervals of time, (e.g., in French) this characteristic is sometimes referred to as isosyllabism or isosyllabicity. However, very little work has been done on the accuracy or general applicability of such properties, and the usefulness of the typology has been questioned. So it is important to remember that the division of languages in two categories (i.e., syllable vs. stress timed languages) is not based on strong concepts. Many phoneticians disagree with the basic idea of timing value. They are of the view that there are three dimensions: fixed word stress (mainly found in Romance languages), variable word stress (mainly found in languages such as English and German) and fixed phrase stress (phrase as a third possibility as exhibited by Japanese) and they want to categorize languages on the basis of these three patterns.

Topic-156: Pitch as a Suprasegmental Feature

As a suprasegmental feature, pitch is an auditory sensation - when we hear a regularly vibrating sound such as a note played on a musical instrument (or a vowel produced by the human voice), we hear a high pitch (when the rate of vibration is high) and a low pitch (when the rate of vibration is low). There

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Further readings on this section:

- Chapter 10 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)
- Chapters 15 to 19 of the additional reading book (**English Phonetics and Phonology-A Practical Course** by Peter Roach)

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SUPRASEGMENTAL FEATURES-III

At the end of this section, the students will be able *to:*

- DEFINE, EXAMINE and EVALUATE suprasegmental features such as intonation, tone and pitch.

Topic-157: Tone and Tonal Languages

Tone (in phonetics and phonology) as a suprasegmental feature refers to an identifiable movement (variation) or level of pitch that is used in a linguistically contrastive way. In tone (tonal) languages, the linguistic function of tone is to change the meaning of a word. For example, in Mandarin Chinese, [ma] said with a high pitch means 'mother' while [ma] said on a low rising tone means 'hemp'. In other (non-tonal) languages, tone forms the central part of intonation, and the difference between, for example, a rising and a falling tone on a particular word may cause a different interpretation of the sentence in which it occurs. In the case of tone languages, it is usual to identify tones as being a property of individual syllables, whereas an intonational tone may be spread over many syllables. In the analysis of English intonation, tone refers to one of the pitch possibilities for the tonic (or nuclear) syllable. For further analysis, a set of four types of tone is usually used (fall, rise, fall–rise and rise–fall) though others are also suggested by various experts.

Topic-158: Intonation as a Suprasegmental Feature

In other (non-tonal) languages, tone forms the central part of intonation, and the difference between, for example, a rising and a falling tone on a particular word may cause a different interpretation of the sentence in which it occurs. In the case of tone languages, it is usual to identify tones as being a property of individual syllables, whereas an intonational tone may be spread over many syllables. In the analysis of English intonation, tone refers to one of the pitch possibilities for the tonic (or nuclear) syllable. For further analysis, a set of four types of tone is usually used (fall, rise, fall–rise and rise–fall) though others are also suggested by various experts.

In simple sense, 'intonation' refers to the variations in the pitch of a speaker's voice used to convey or alter meaning (at sentence level). In its broader and more popular sense, it is used to cover much the same field as 'prosody', where various features such as voice quality, tempo and loudness are also included. It is a term frequently used in the study of suprasegmental phonology, referring to the distinctive use of patterns of pitch, or melody and the study of intonation is sometimes called intonology. Experts have suggested several ways of analyzing intonation. In some approaches, the pitch patterns are described as contours and analyzed in terms of levels of pitch as pitch phonemes and morphemes while in others, the patterns are described as tone units or tone groups which are further analyzed as contrasts of nuclear tone, tonicity, etc. The three variables of pitch range, height and direction are generally distinguished. Some approaches, especially within pragmatics, operate with a much broader notion than that of the tone unit: intonational phrasing is a structured hierarchy of the intonational constituents in conversation. A formal category of intonational phrase is also sometimes recognized which is an utterance span dominated by boundary tones.

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Topic-159: Functions of Intonations

Intonation as a suprasegmental feature performs several functions in a language. Its most important function is to act as a signal of grammatical structure (e.g., creating patterns to distinguish among grammatical categories), where it performs a role similar to punctuation (in written language). It

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may furnish far more contrasts (for conveying meaning). Intonation also gives an idea about the syntactic boundaries (sentence, clause and phrase level boundaries). It also provides the contrast between some grammatical structures (such as questions and statements). For example, the change in meaning illustrated

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the communication is quite important as it also conveys personal attitude (e.g., sarcasm, puzzlement, anger, etc.). Finally, it can signal contrasts in pitch along with other prosodic and paralinguistic features. It can also bring variation in meaning and can prove an important signal of the social background of the speakers.

Topic-160: Explaining Functions of Intonation

While discussing the functions of intonation, another approach is to concentrate on its role in conversational discourse. This involves such aspects as indicating whether the particular thing being said constitutes new information or old (in sentence, for example). It further creates the regulation of turn-taking in conversation, the establishment of dominance and the elicitation of co-operative responses as well. As with the signaling of attitudes, it seems that though analysts concentrate on pitch movements there are many other prosodic factors being used to create these effects. It is also important to note here that much less work has been done on the intonation of languages other than English. It seems that all languages have something that can be identified as intonation (there can be many differences between among languages). It is, therefore, a potential area to be studied and explored further. One reason for intonation not being well explored is because of the different descriptive frameworks used by different analysts for studying intonation cross-linguistically. Even it is claimed that tone languages also have intonation and that it is superimposed upon the tones themselves. Such a claim creates especially difficult problems of analysis.

Further readings on this section:

- Chapter 10 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)

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- DEFINE and EXPLAIN linguistic phonetics in the broader perspective of IPA.

Topic-161: Linguistic Phonetics

Linguistic phonetics is an approach which is embodied in the principles of the International Phonetic Association (IPA) and in a hierarchical phonetic descriptive framework that provides certain basis for formal phonological theory. Speech, being very complex phenomena and having multiple levels of organization, needs to be explored from different angles. Linguistic phonetics answers a number of questions related to the possible ways of articulatory unified phonetics and phonology and from the perspective of cognitive phonetics focusing on speech production and perception and how they shape languages as a sound systems. The idea is mainly related to the overall ability of human beings to produce sounds (as a community and irrespective of their specific languages) and then the representation of their shared knowledge (as considered by the IPA in its charts) for formal phonetic and phonological theories.

Topic-162: Phonetics of the Community and of the Individual

Linguistic phonetic descriptions (of speech sounds) are, by and large, descriptions of the phonetics of the community (excluding the individual properties and considering the shared properties of the sound system of a language by its native speakers). The representations that experts write and use in the IPA, and analyze in a formal phonological theory, are intended to show the community's shared knowledge of how to say the words of a language. It is important to note that this shared phonetic knowledge is perceptible to other speakers (and thus to the phonetician as well) and experts are mainly related to the aggregate behavior of linguistic group in the sense that it captures what community members accept as correct pronunciation system. So the focus of linguistic phonetic description is the

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Topic-163: Explaining Phonetics of the Community and of the Individual

The major reason why the phonetics of the community is considered for phonetic descriptions is that, firstly, individual speakers differ in interesting ways (two native speakers of a language will always speak with some variations). The description of the phonetics of the individual involves describing the phonetic knowledge and skills related to the performance of language. It is possible that certain aspects of the phonetics of the individual can be captured using IPA transcription but others are not compatible with it (such as his private knowledge and its performance and the role of memory and experience). Secondly, the phonetics of the individual is usually not the focus of the linguist in speech elicitation, and it is

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While discussing the key elements of linguistic phonetic description, we need to consider the International Phonetic Alphabet (abbreviated as IPA). IPA is the set of symbols and diacritics that have been officially approved by IPA. The association publishes a chart comprising of a number of separate charts. At the top inside the front cover, you will find the main consonant chart. Below it is a table showing the symbols for nonpulmonic consonants, and below that is the vowel chart. Inside the back cover is a list of diacritics and other symbols, and a set of symbols for suprasegmental features (events) such as tone, intonation, stress, and length. Remember that the IPA chart does not try to cover all possible types of phonetic descriptions (e.g., all the individual strategies for realizing linguistic phonological contrasts, or gradations in the degree of co-articulation between adjacent segments, etc.). Instead, it is limited to those possible sounds that can have linguistic significance in that they can change the meaning of a word in some languages. So the description of IPA is based on the linguistic phonetics of the community.

Topic-165: Explaining the IPA

One evidence that the IPA chart is based on linguistic phonetics is the description of the blank cells on the chart (those neither shaded nor containing a symbol) that indicate the combinations of categories that are humanly possible but have not been observed so far to be distinctive in any language (e.g., a voiceless retroflex lateral fricative is possible but has not been documented so far, so it is left blank). The shaded cells, on the other hand, exhibit the sounds not possible at these places. Further, below the consonant chart is a set of symbols for consonants made with different airstream mechanisms (clicks, voiced implosives, and ejectives). All these descriptions reflect the potentialities of human speech sounds

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Further readings on this section:

- Chapter 11 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)

Lesson-33

LINGUISTIC PHONETICS-II

At the end of this section, the students will be able to:

- DEFINE and COMMENT on 'feature hierarchy' and problems with linguistic explanations.

Topic-166: Feature Hierarchy

The second way by which we will consider why the linguistic phonetic description is important is by considering a hierarchical organization of linguistic phonetic features. Feature hierarchy is an important concept in phonetics and phonology which is based on the properties and features of sounds. In a very general sense, a feature may be tied to a particular articulatory maneuver or acoustic property. For example, the feature [bilabial] indicates not only that the segment is produced with lips but also that it involves both of them. Such features (in phonetics and phonology) are listed in a hierarchy with nodes in the hierarchy defining ever more specific phonetic properties. For example, sounds are divided in terms of their supra-laryngeal and laryngeal characteristics, and their airstream mechanism. The supra-laryngeal characteristics can be further divided into those for place (of articulation), manner (of articulation), the possibility of nasality, and the possibility of being lateral. Thus, these features are used for classifying speech sounds and describing them formally.

Topic-167: Feature Hierarchy: Discussion

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Similarly, 'Approximant' and 'Vowel' have five principal features: Height, (with five possible values [high], [mid-high], [mid], [mid-low], and [low]), Backness (with three values, [front], [center], and [back]) and two kinds of 'Rounding' (i.e., Protrusion with possible values [protruded] and [retracted], and Compression, with possible values [compressed] and [separated]). The fourth feature for vowels and approximants is the 'Tongue Root' which has two possible values: [+ATR] and [−ATR]. Finally, the feature 'Rhotic' has only one possible value, [rhotacized]. It is also important to remember that the 'Laryngeal' possibilities involve mainly five features ([voiceless], [breathy voice], [modal voice], [creaky voice], and [closed] - forming a glottal stop). Airstream features have three values; Pulmonic, Velaric and Glottalic).

Topic-169: Problems with Linguistic Explanations

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explanations that are restricted in this way often fall into the fallacy of reification (i.e., acting as if abstract things are concrete).

Topic-170: Explaining the Problem with Linguistic Explanations

The phonetics of the individual gives explanation to a number of such complex issues. For example, take the case of 'assimilation' - it happens because there is a tendency in pronunciation for adjacent sounds to share phonetic properties (at individual level). This "explanation" is even more impressive if we state it as a formal constraint (in Optimality theory) on sequences of sounds as the following:

AGREE(x): Adjacent output segments have the same value of the feature x.

It means that assimilation exists when adjacent segments share features. In simple words we can say that the adjacent sounds would influence the sounds in the neighborhood and that the tendency to assimilate (a cross-linguistic generalization) exists because there is a tendency to assimilate (reified as a specific "explanatory principle"). The private phonetic knowledge of the individual answers for a more satisfying way to explain language sound patterns.

Further readings on this section:

- Chapter 11 of the textbook (**A Course in Phonetics** by Peter Ladefoged and Keith Johnson)

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into three mutually exclusive possibilities: Laminal (i.e., blade of the tongue), Apical (i.e., tip of the tongue), and Sub-apical (i.e., the under part of the blade of the tongue). Thus the major regions may be subdivided into sub regions on the basis of their features.

Topic-168: Feature Hierarchy: Explanation

This grouping under feature hierarchy also reflects the fact that the four features (i.e., Stop, Fricative, Approximant, and Vowel) depend on the degree of closure of the articulators. In order to fully grasp the feature hierarchy, we need to explore the major divisions in this topic. The manner category 'Stop' has only one possible value (i.e., [stop]), but 'Fricative' has two (i.e., [sibilant] and [nonsibilant]).

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Similarly, 'Approximant' and 'Vowel' have five principal features: Height, (with five possible values [high], [mid-high], [mid], [mid-low], and [low]), Backness (with three values, [front], [center], and [back]) and two kinds of 'Rounding' (i.e., Protrusion with possible values [protruded] and [retracted], and

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Topic-170: Explaining the Problem with Linguistic Explanations

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phenomenon of 'assimilation' where the adjacent sounds come to share some phonetic properties. But if we restrict ourselves to the terminology and knowledge base of linguistic phonetics (the phonetics of the community); we are restricted to descriptions of sound patterns and not their explanation. In fact, explanations that are restricted in this way often fall into the fallacy of reification (i.e., acting as if abstract things are concrete).

Topic-170: Explaining the Problem with Linguistic Explanations

The phonetics of the individual gives explanation to a number of such complex issues. For example, take the case of 'assimilation' - it happens because there is a tendency in pronunciation for adjacent sounds to share phonetic properties (at individual level). This "explanation" is even more impressive if we state it as a formal constraint (in Optimality theory) on sequences of sounds as the following:

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